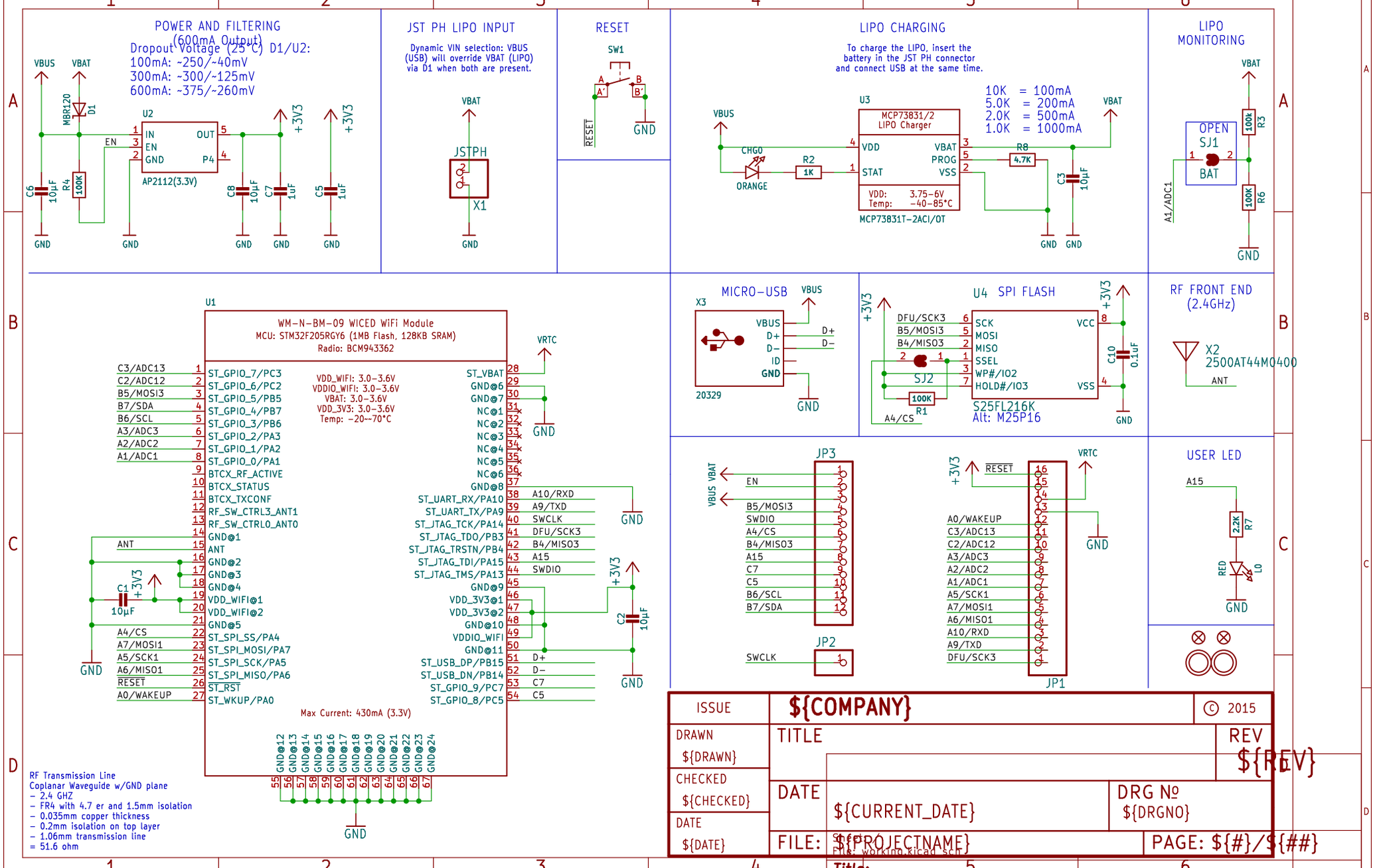
POWER AND FILTERING
(600mA Output)
Dropout Voltage (25°C) D1/U2:
100mA: ~250/~40mV
300mA: ~300/~125mV
600mA: ~375/~260mV
VBUS
VBAT
MBR120
D1
U2
IN
EN
GND
OUT
P4
AP2112(3.3V)
R4
100K
C6
10µF
C8
10µF
C7
1uF
C5
1uF
+3V3
JST PH LIPO INPUT
Dynamic VIN selection: VBUS (USB) will override VBAT (LIPO) via D1 when both are present.
JSTPH
X1
RESET
SW1
RESET
LIPO CHARGING
To charge the LIPO, insert the battery in the JST PH connector and connect USB at the same time.
U3
MCP73831/2
LIPO Charger
VDD
VBAT
PROG
VSS
STAT
CHGO
ORANGE
R2
1K
R8
4.7K
C3
10µF
VDD: 3.75-6V
Temp: -40-85°C
MCP73831T-2ACI/OT
10K = 100mA
5.0K = 200mA
2.0K = 500mA
1.0K = 1000mA
LIPO MONITORING
OPEN
SJ1
BAT
R3
100K
R6
100K
A1/ADC1
U1
WM-N-BM-09 WICED WiFi Module
MCU: STM32F205RGY6 (1MB Flash, 128KB SRAM)
Radio: BCM943362
VDD_WIFI: 3.0-3.6V
VDDIO_WIFI: 3.0-3.6V
VBAT: 3.0-3.6V
VDD_3V3: 3.0-3.6V
Temp: -20~+70°C
C3/ADC13
C2/ADC12
B5/MOSI3
B7/SDA
B6/SCL
A3/ADC3
A2/ADC2
A1/ADC1
ST_GPIO_7/PC3
ST_GPIO_6/PC2
ST_GPIO_5/PB5
ST_GPIO_4/PB7
ST_GPIO_3/PB6
ST_GPIO_2/PA3
ST_GPIO_1/PA2
ST_GPIO_0/PA1
BTCX_RF_ACTIVE
BTCX_STATUS
BTCX_TXCONF
RF_SW_CTRL3_ANT1
RF_SW_CTRL0_ANT0
GND@1
ANT
GND@2
GND@3
GND@4
VDD_WIFI@1
VDD_WIFI@2
GND@5
A4/CS
A7/MOSI1
A5/SCK1
A6/MISO1
RESET
A0/WAKEUP
ST_SPI_SS/PA4
ST_SPI_MOSI/PA7
ST_SPI_SCK/PA5
ST_SPI_MISO/PA6
ST_RST
ST_WKUP/PA0
ST_VBAT
GND@6
GND@7
NC@1
NC@2
NC@3
NC@4
NC@5
NC@6
GND@8
ST_UART_RX/PA10
ST_UART_TX/PA9
ST_JTAG_TCK/PA14
ST_JTAG_TDO/PB3
ST_JTAG_TRSTN/PB4
ST_JTAG_TDI/PA15
ST_JTAG_TMS/PA13
GND@9
VDD_3V3@1
VDD_3V3@2
GND@10
VDDIO_WIFI
GND@11
ST_USB_DP/PB15
ST_USB_DN/PB14
ST_GPIO_9/PC7
ST_GPIO_8/PC5
A10/RXD
A9/TXD
SWCLK
DFU/SCK3
B4/MISO3
A15
SWDIO
D+
D-
C7
C5
Max Current: 430mA (3.3V)
GND@12
GND@13
GND@14
GND@15
GND@16
GND@17
GND@18
GND@19
GND@20
GND@21
GND@22
GND@23
GND@24
C1
10µF
C2
10µF
VRTC
RF Transmission Line
Coplanar Waveguide w/GND plane
- 2.4 GHZ
- FR4 with 4.7 er and 1.5mm isolation
- 0.035mm copper thickness
- 0.2mm isolation on top layer
- 1.06mm transmission line
= 51.6 ohm
MICRO-USB
X3
VBUS
D+
D-
ID
GND
20329
U4 SPI FLASH
DFU/SCK3
B5/MOSI3
B4/MISO3
SCK
MOSI
MISO
SSEL
WP#/IO2
HOLD#/IO3
VCC
VSS
SJ2
100K
R1
A4/CS
C10
0.1uF
S25FL216K
Alt: M25P16
RF FRONT END
(2.4GHz)
X2
2500AT44M0400
ANT
JP3
JP2
JP1
SWCLK
USER LED
A15
2.2K
R7
RED
L0
ISSUE
${COMPANY}
© 2015
DRAWN
${DRAWN}
TITLE
REV
${REV}
CHECKED
${CHECKED}
DATE
${DATE}
${CURRENT_DATE}
DRG Nº
${DRGNO}
FILE: ${PROJECTNAME}
Sheet: /
File: working.kicad_sch
PAGE: ${#}/${##}
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1